THURSDAY 12 NOVEMBER 2026 · 09:00–10:00 · GROUP 07

Introduction to Pure Mathematics — Week 8 tutorial

Permutations in cycle notation, calculating in dihedral groups, cancelling in a group

1 · Example
Watch one worked example and copy it down.

2 · Your turn
Do one like it with your neighbour.

3 · Compare
We check it together; write one takeaway.

Key ideas

1. **Cycles.** $(1, 5, 3, 7)$ means $1 \mapsto 5 \mapsto 3 \mapsto 7 \mapsto 1$; numbers not written stay fixed. *Recipe:* start at the smallest number, follow it until you return, close the bracket, restart at the smallest unused number.
2. **Compose right to left.** fg means "g first, then f": $(fg)(k) = f(g(k))$. Usually $fg \neq gf$.
3. **Dihedral group.** $D_{2n} = \{e, a, \ldots, a^{n-1}, b, ab, \ldots, a^{n-1}b\}$ with rules $a^n = e,\ \ b^2 = e,\ \ ba^i = a^{-i}b$.
4. **Cancel on the same side.** $xz = yz \Rightarrow x = y$ and $zx = zy \Rightarrow x = y$. Left undoes left, right undoes right.

Recall

No notes, 2 minutes on your own, then compare with your neighbour.

1. Which of these is **not** a group, and which axiom fails? $(\mathbb{Z}, +),\ \ (\mathbb{Z} \setminus \{0\}, \times),\ \ (\mathbb{R} \setminus \{0\}, \times)$.

2. In a group G, solve $ax = b$ for x.

3. True or false: in every group, $xy = yx$.

After the tutorial

- **From today:** finish A7(b); then B8 and C1 (they use only the group axioms).
- **After this week's lectures on order and subgroups:** A4, A5, A6, A8, then B1.
- **Assigned Question (ASSESSED):** due 13:00 Thursday 19 November. Today's Rounds 1 and 2 are the methods you need to start it.

Solutions to all Part A questions are on Blackboard now.

Adapted from Introduction to Pure Mathematics Problem Sheet 7 (University of Bristol, 2026/27) for tutorial use. No solutions are included.

Round 1 · Permutations: from the array to cycles, and composing

EXAMPLE — we do this one together (Pure PS7 A2)

Express the following permutation as a product of disjoint cycles:

$$g = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 5 & 4 & 7 & 6 & 3 & 2 & 1 \end{pmatrix} \in S_7.$$

YOUR TURN — with your neighbour (Pure PS7 A1)

Let

$$f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 4 & 1 & 3 & 2 \end{pmatrix}, \qquad g = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 4 & 1 & 5 \end{pmatrix}.$$

Fill in the table (remember: g first, then f), then write fg as a product of disjoint cycles.

k	1	2	3	4	5
$g(k)$					
$f(g(k))$					

$fg =$

Finished early? Compute gf as well. Is it the same as fg?

Takeaway — in your own words:

Challenge — harder; work on the back of this page (Pure PS7 C4(a)(b))

(a) Check that $(1,2,3,4) = (1,2)(2,3)(3,4)$. Explain why *every* permutation in S_n can be written as a product of 2-cycles.

(b) Write $(1,3)$ as a product of three 2-cycles of the form $(i, i+1)$. Can you do the same for $(1,4)$? How many do you need?

Round 2 · Dihedral groups: push every b to the right

EXAMPLE — we do this one together (Example in D_{12} (hexagon), same method as PS7 A3)

In $D_{12} = \{e, a, \ldots, a^5, b, ab, \ldots, a^5b\}$ (the symmetries of a regular hexagon), write each of these in the form a^ib^j with $0 \leqslant i \leqslant 5$, $j \in \{0, 1\}$:

(i) $ba^2\,ba^5$ (ii) $a^4\,ba^3$.

YOUR TURN — with your neighbour (Pure PS7 A3)

Let $G = D_{20} = \{e, a, \ldots, a^9, b, ab, \ldots, a^9b\}$. Express the element

$$b^3a^5ba^{-7}$$

in the form a^ib^j with $i \in \{0, 1, \ldots, 9\}$ and $j \in \{0, 1\}$.

Rules in D_{20}: $a^{\square} = e$, $b^{\square} = e$, $ba^i = _____$

Finished early? Simplify $(a^3b)^2$ in D_{20}. What does the answer tell you about the symmetry a^3b?

Takeaway — in your own words:

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(a) Show that in D_{2n} the product of two reflections is a rotation: simplify $a^ib \cdot a^jb$.

(b) In D_8 (the square), find *all* elements x such that $xy = yx$ for every $y \in D_8$. What are these symmetries geometrically?

Round 3 · Reasoning in any group: cancel on the same side

if time

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Let G be a group. For fixed $a, b, c \in G$, prove that the equation $axb = c$ has a unique solution, and find it.

YOUR TURN — with your neighbour (Pure PS7 A7(c))

A student argues: "If $xz = zy$, we can cancel z, so $x = y$."

(i) What is wrong with this argument?

(ii) Test it in S_3 with $x = (1,2)$, $y = (1,3)$, $z = (1,2,3)$: compute xz and zy.

*Finished early? Prove A7(b): $(xy)^2 = x^2y^2 \iff xy = yx$. **Challenge (C1):** if $x^2 = e$ for every $x \in G$, prove that G is abelian.*

Takeaway — in your own words:

Exit ticket · on your own, 3 min

Least sure about? Circle: Round 1 | Round 2 | Round 3

1. Write $(1,2,3)(1,2)$ as a product of disjoint cycles.
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in the form a^ib^j with $i \in \{0, 1, \ldots, 9\}$ and $j \in \{0, 1\}$.

Rules in D_{20}: $a^{\square} = e$, $b^{\square} = e$, $ba^i = _____$

Finished early? Simplify $(a^3b)^2$ in D_{20}. What does the answer tell you about the symmetry a^3b?

Takeaway — in your own words:

Challenge — harder; work on the back of this page

(a) Show that in D_{2n} the product of two reflections is a rotation: simplify $a^ib \cdot a^jb$.

(b) In D_8 (the square), find *all* elements x such that $xy = yx$ for every $y \in D_8$. What are these symmetries geometrically?

Round 3 · Reasoning in any group: cancel on the same side

if time

EXAMPLE — we do this one together (Pure PS7 A7(a))

Let G be a group. For fixed $a, b, c \in G$, prove that the equation $axb = c$ has a unique solution, and find it.

YOUR TURN — with your neighbour (Pure PS7 A7(c))

A student argues: "If $xz = zy$, we can cancel z, so $x = y$."

(i) What is wrong with this argument?

(ii) Test it in S_3 with $x = (1,2)$, $y = (1,3)$, $z = (1,2,3)$: compute xz and zy.

*Finished early? Prove A7(b): $(xy)^2 = x^2y^2 \iff xy = yx$. **Challenge (C1):** if $x^2 = e$ for every $x \in G$, prove that G is abelian.*

Takeaway — in your own words:

Exit ticket · on your own, 3 min

Least sure about? Circle: Round 1 | Round 2 | Round 3

1. Write $(1,2,3)(1,2)$ as a product of disjoint cycles.
2. In D_{2n}, simplify bab.
3. In a group, solve $xa = b$ for x.

THURSDAY 12 NOVEMBER 2026 · 09:00–10:00 · GROUP 07

Introduction to Pure Mathematics — Week 8 tutorial

Permutations in cycle notation, calculating in dihedral groups, cancelling in a group

1 · Example
Watch one worked example and copy it down.

2 · Your turn
Do one like it with your neighbour.

3 · Compare
We check it together; write one takeaway.

Key ideas

1. **Cycles.** $(1, 5, 3, 7)$ means $1 \mapsto 5 \mapsto 3 \mapsto 7 \mapsto 1$; numbers not written stay fixed. *Recipe:* start at the smallest number, follow it until you return, close the bracket, restart at the smallest unused number.
2. **Compose right to left.** fg means "g first, then f": $(fg)(k) = f(g(k))$. Usually $fg \neq gf$.
3. **Dihedral group.** $D_{2n} = \{e, a, \ldots, a^{n-1}, b, ab, \ldots, a^{n-1}b\}$ with rules $a^n = e,\ \ b^2 = e,\ \ ba^i = a^{-i}b$.
4. **Cancel on the same side.** $xz = yz \Rightarrow x = y$ and $zx = zy \Rightarrow x = y$. Left undoes left, right undoes right.

Recall

No notes, 2 minutes on your own, then compare with your neighbour.

1. Which of these is **not** a group, and which axiom fails? $(\mathbb{Z}, +),\ \ (\mathbb{Z} \setminus \{0\}, \times),\ \ (\mathbb{R} \setminus \{0\}, \times)$.

2. In a group G, solve $ax = b$ for x.

3. True or false: in every group, $xy = yx$.

After the tutorial

- **From today:** finish A7(b); then B8 and C1 (they use only the group axioms).
- **After this week's lectures on order and subgroups:** A4, A5, A6, A8, then B1.
- **Assigned Question (ASSESSED):** due 13:00 Thursday 19 November. Today's Rounds 1 and 2 are the methods you need to start it.

Solutions to all Part A questions are on Blackboard now.

Adapted from Introduction to Pure Mathematics Problem Sheet 7 (University of Bristol, 2026/27) for tutorial use. No solutions are included.

Round 1 · Permutations: from the array to cycles, and composing

EXAMPLE — we do this one together (Pure PS7 A2)

Express the following permutation as a product of disjoint cycles:

$$g = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 5 & 4 & 7 & 6 & 3 & 2 & 1 \end{pmatrix} \in S_7.$$

YOUR TURN — with your neighbour (Pure PS7 A1)

Let

$$f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 4 & 1 & 3 & 2 \end{pmatrix}, \qquad g = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 4 & 1 & 5 \end{pmatrix}.$$

Fill in the table (remember: g first, then f), then write fg as a product of disjoint cycles.

k	1	2	3	4	5
$g(k)$					
$f(g(k))$					

$fg =$

Finished early? Compute gf as well. Is it the same as fg?

Takeaway — in your own words:

Challenge — harder; work on the back of this page (Pure PS7 C4(a)(b))

(a) Check that $(1,2,3,4) = (1,2)(2,3)(3,4)$. Explain why *every* permutation in S_n can be written as a product of 2-cycles.

(b) Write $(1,3)$ as a product of three 2-cycles of the form $(i, i+1)$. Can you do the same for $(1,4)$? How many do you need?

Round 2 · Dihedral groups: push every b to the right

EXAMPLE — we do this one together (Example in D_{12} (hexagon), same method as PS7 A3)

In $D_{12} = \{e, a, \ldots, a^5, b, ab, \ldots, a^5b\}$ (the symmetries of a regular hexagon), write each of these in the form a^ib^j with $0 \leqslant i \leqslant 5$, $j \in \{0, 1\}$:

(i) $ba^2\,ba^5$ (ii) $a^4\,ba^3$.

YOUR TURN — with your neighbour (Pure PS7 A3)

Let $G = D_{20} = \{e, a, \ldots, a^9, b, ab, \ldots, a^9b\}$. Express the element

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if time

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Round 1 · Permutations: from the array to cycles, and composing

EXAMPLE — we do this one together (Pure PS7 A2)

Express the following permutation as a product of disjoint cycles:

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Let

$$f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 4 & 1 & 3 & 2 \end{pmatrix}, \qquad g = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 4 & 1 & 5 \end{pmatrix}.$$

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Finished early? Compute gf as well. Is it the same as fg?

Takeaway — in your own words:

Challenge — harder; work on the back of this page (Pure PS7 C4(a)(b))

(a) Check that $(1,2,3,4) = (1,2)(2,3)(3,4)$. Explain why *every* permutation in S_n can be written as a product of 2-cycles.

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Round 2 · Dihedral groups: push every b to the right

EXAMPLE — we do this one together (Example in D_{12} (hexagon), same method as PS7 A3)

In $D_{12} = \{e, a, \ldots, a^5, b, ab, \ldots, a^5b\}$ (the symmetries of a regular hexagon), write each of these in the form a^ib^j with $0 \leqslant i \leqslant 5$, $j \in \{0, 1\}$:

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Let $G = D_{20} = \{e, a, \ldots, a^9, b, ab, \ldots, a^9b\}$. Express the element

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Round 3 · Reasoning in any group: cancel on the same side

if time

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Let G be a group. For fixed $a, b, c \in G$, prove that the equation $axb = c$ has a unique solution, and find it.

YOUR TURN — with your neighbour (Pure PS7 A7(c))

A student argues: "If $xz = zy$, we can cancel z, so $x = y$."

(i) What is wrong with this argument?

(ii) Test it in S_3 with $x = (1,2)$, $y = (1,3)$, $z = (1,2,3)$: compute xz and zy.

*Finished early? Prove A7(b): $(xy)^2 = x^2y^2 \iff xy = yx$. **Challenge (C1):** if $x^2 = e$ for every $x \in G$, prove that G is abelian.*

Takeaway — in your own words:

Exit ticket · on your own, 3 min

Least sure about? Circle: Round 1 | Round 2 | Round 3

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THURSDAY 12 NOVEMBER 2026 · 09:00–10:00 · GROUP 07

Introduction to Pure Mathematics — Week 8 tutorial

Permutations in cycle notation, calculating in dihedral groups, cancelling in a group

1 · Example
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Key ideas

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- **From today:** finish A7(b); then B8 and C1 (they use only the group axioms).
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Round 1 · Permutations: from the array to cycles, and composing

EXAMPLE — we do this one together (Pure PS7 A2)

Express the following permutation as a product of disjoint cycles:

$$g = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 5 & 4 & 7 & 6 & 3 & 2 & 1 \end{pmatrix} \in S_7.$$

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Let

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Finished early? Compute gf as well. Is it the same as fg?

Takeaway — in your own words:

Challenge — harder; work on the back of this page (Pure PS7 C4(a)(b))

(a) Check that $(1,2,3,4) = (1,2)(2,3)(3,4)$. Explain why *every* permutation in S_n can be written as a product of 2-cycles.

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Round 2 · Dihedral groups: push every b to the right

EXAMPLE — we do this one together (Example in D_{12} (hexagon), same method as PS7 A3)

In $D_{12} = \{e, a, \ldots, a^5, b, ab, \ldots, a^5b\}$ (the symmetries of a regular hexagon), write each of these in the form a^ib^j with $0 \leqslant i \leqslant 5$, $j \in \{0, 1\}$:

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Round 3 · Reasoning in any group: cancel on the same side

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Takeaway — in your own words:

Exit ticket · on your own, 3 min

Least sure about? Circle: Round 1 | Round 2 | Round 3

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Introduction to Pure Mathematics — Week 8 tutorial

Permutations in cycle notation, calculating in dihedral groups, cancelling in a group

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After the tutorial

- **From today:** finish A7(b); then B8 and C1 (they use only the group axioms).
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EXAMPLE — we do this one together (Pure PS7 A2)

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Round 1 · Permutations: from the array to cycles, and composing

EXAMPLE — we do this one together (Pure PS7 A2)

Express the following permutation as a product of disjoint cycles:

$$g = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 5 & 4 & 7 & 6 & 3 & 2 & 1 \end{pmatrix} \in S_7.$$

YOUR TURN — with your neighbour (Pure PS7 A1)

Let

$$f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 4 & 1 & 3 & 2 \end{pmatrix}, \qquad g = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 4 & 1 & 5 \end{pmatrix}.$$

Fill in the table (remember: g first, then f), then write fg as a product of disjoint cycles.

k	1	2	3	4	5
$g(k)$					
$f(g(k))$					

$fg =$

Finished early? Compute gf as well. Is it the same as fg?

Takeaway — in your own words:

Challenge — harder; work on the back of this page (Pure PS7 C4(a)(b))

(a) Check that $(1,2,3,4) = (1,2)(2,3)(3,4)$. Explain why *every* permutation in S_n can be written as a product of 2-cycles.

(b) Write $(1,3)$ as a product of three 2-cycles of the form $(i, i+1)$. Can you do the same for $(1,4)$? How many do you need?

Round 2 · Dihedral groups: push every b to the right

EXAMPLE — we do this one together (Example in D_{12} (hexagon), same method as PS7 A3)

In $D_{12} = \{e, a, \ldots, a^5, b, ab, \ldots, a^5b\}$ (the symmetries of a regular hexagon), write each of these in the form a^ib^j with $0 \leqslant i \leqslant 5$, $j \in \{0, 1\}$:

(i) $ba^2\,ba^5$ (ii) $a^4\,ba^3$.

YOUR TURN — with your neighbour (Pure PS7 A3)

Let $G = D_{20} = \{e, a, \ldots, a^9, b, ab, \ldots, a^9b\}$. Express the element

$$b^3a^5ba^{-7}$$

in the form a^ib^j with $i \in \{0, 1, \ldots, 9\}$ and $j \in \{0, 1\}$.

Rules in D_{20}: $a^{\square} = e$, $b^{\square} = e$, $ba^i = _____$

Finished early? Simplify $(a^3b)^2$ in D_{20}. What does the answer tell you about the symmetry a^3b?

Takeaway — in your own words:

Challenge — harder; work on the back of this page

(a) Show that in D_{2n} the product of two reflections is a rotation: simplify $a^ib \cdot a^jb$.

(b) In D_8 (the square), find *all* elements x such that $xy = yx$ for every $y \in D_8$. What are these symmetries geometrically?

Round 3 · Reasoning in any group: cancel on the same side

if time

EXAMPLE — we do this one together (Pure PS7 A7(a))

Let G be a group. For fixed $a, b, c \in G$, prove that the equation $axb = c$ has a unique solution, and find it.

YOUR TURN — with your neighbour (Pure PS7 A7(c))

A student argues: "If $xz = zy$, we can cancel z, so $x = y$."

(i) What is wrong with this argument?

(ii) Test it in S_3 with $x = (1,2)$, $y = (1,3)$, $z = (1,2,3)$: compute xz and zy.

*Finished early? Prove A7(b): $(xy)^2 = x^2y^2 \iff xy = yx$. **Challenge (C1):** if $x^2 = e$ for every $x \in G$, prove that G is abelian.*

Takeaway — in your own words:

Exit ticket · on your own, 3 min

Least sure about? Circle: Round 1 | Round 2 | Round 3

1. Write $(1,2,3)(1,2)$ as a product of disjoint cycles.
2. In D_{2n}, simplify bab.
3. In a group, solve $xa = b$ for x.